

Problem 1. SVM.

In this problem, you will prove that the hard-SVM rule, namely,

$$\operatorname{argmax}_{(w,b): \|w\|=1} \min_{i \in [m]} |\langle w, x_i \rangle + b| \quad \text{s.t.} \quad \forall i, \ y_i(\langle w, x_i \rangle + b) > 0,$$

is equivalent to the following formulation:

$$\operatorname{argmax}_{(w,b): \|w\|=1} \min_{i \in [m]} y_i(\langle w, x_i \rangle + b).$$

1. Define $\mathcal{G} := \{(w, b) : \forall i, \ y_i(\langle w, x_i \rangle + b) > 0\}$. Show that

$$\operatorname{argmax}_{(w,b): \|w\|=1} \min_{i \in [m]} y_i(\langle w, x_i \rangle + b) \in \mathcal{G}.$$

2. Show that for any $(w, b) \in \mathcal{G}$,

$$\min_{i \in [m]} y_i(\langle w, x_i \rangle + b) = \min_{i \in [m]} |\langle w, x_i \rangle + b|,$$

and finish the proof of the equivalence.

Solution.

1. Write

$$\gamma(w, b) := \min_{i \in [m]} y_i(\langle w, x_i \rangle + b).$$

By linear separability, there is a separating pair $(\bar{w}, \bar{b})$ with $\|\bar{w}\| = 1$: simply normalize any separating pair by dividing both w and b by $\|w\|$. Since the sample is finite,

$$\gamma(\bar{w}, \bar{b}) > 0.$$

Therefore, every maximizer (w^*, b^*) of the second formulation satisfies

$$\min_{i \in [m]} y_i(\langle w^*, x_i \rangle + b^*) = \gamma(w^*, b^*) \geq \gamma(\bar{w}, \bar{b}) > 0.$$

Thus $y_i(\langle w^*, x_i \rangle + b^*) > 0$ for each $i \in [m]$, which is exactly

$$(w^*, b^*) \in \mathcal{G}.$$

2. For any $(w, b) \in \mathcal{G}$, the number $\langle w, x_i \rangle + b$ has sign y_i . Consequently,

$$y_i(\langle w, x_i \rangle + b) = |\langle w, x_i \rangle + b| \quad \text{for every } i,$$

and taking the minimum gives the desired equality. Part 1 shows that restricting the second optimization to $\mathcal{G}$ removes no maximizers. Part 2 shows that, on this restricted domain, its objective is exactly the first objective. Hence

$$\begin{aligned} \operatorname{argmax}_{\|w\|=1} \min_i y_i(\langle w, x_i \rangle + b) &= \operatorname{argmax}_{\substack{\|w\|=1 \\ (w,b) \in \mathcal{G}}} \min_i y_i(\langle w, x_i \rangle + b) \\ &= \operatorname{argmax}_{\substack{\|w\|=1 \\ (w,b) \in \mathcal{G}}} \min_i |\langle w, x_i \rangle + b|. \end{aligned}$$

Problem 2. Rademacher Complexity.

In this problem, you will prove the Massart lemma:

Lemma 1 (Massart lemma). *Let $A = \{a_1, \dots, a_N\}$ be a finite set of vectors in $\mathbb{R}^m$. Define $\bar{a} = \frac{1}{N} \sum_i a_i$. Then, the Rademacher complexity of A can be bounded by:*

$$R(A) \leq \max_{i \in [N]} \|a_i - \bar{a}\| \cdot \frac{\sqrt{2 \log N}}{m}.$$

This is Lemma 26.8 in Understanding Machine Learning.

Solution. Let $\sigma_1, \dots, \sigma_m$ be independent Rademacher random variables, meaning $\mathbb{P}(\sigma_j = 1) = \mathbb{P}(\sigma_j = -1) = 1/2$, and put $\sigma = (\sigma_1, \dots, \sigma_m)$. We use the definition

$$R(A) := \frac{1}{m} \mathbb{E}_\sigma \left[\max_{i \in [N]} \langle \sigma, a_i \rangle \right].$$

Step 1: Centering does not change the complexity. For any fixed vector $c \in \mathbb{R}^m$, we have, for each realization of σ ,

$$\max_{i \in [N]} \langle \sigma, a_i - c \rangle = \max_{i \in [N]} \langle \sigma, a_i \rangle - \langle \sigma, c \rangle.$$

Since $\mathbb{E}[\sigma_j] = 0$ for every j , $\mathbb{E}\langle \sigma, c \rangle = 0$. Taking expectations therefore shows that translating every vector by the same amount leaves $R(A)$ unchanged. In particular, define

$$v_i := a_i - \bar{a}, \quad r := \max_{i \in [N]} \|v_i\|_2, \quad Z_i := \langle \sigma, v_i \rangle.$$

Then

$$m R(A) = \mathbb{E} \left[\max_{i \in [N]} Z_i \right].$$

Step 2: Bound the exponential moment of each Z_i . Fix $t > 0$. Independence of the signs gives

$$\begin{aligned} \mathbb{E} e^{tZ_i} &= \mathbb{E} \prod_{j=1}^m e^{t\sigma_j(v_i)_j} \\ &= \prod_{j=1}^m \mathbb{E} e^{t\sigma_j(v_i)_j} \\ &= \prod_{j=1}^m \frac{e^{t(v_i)_j} + e^{-t(v_i)_j}}{2} = \prod_{j=1}^m \cosh(t(v_i)_j). \end{aligned}$$

For every real number u , the power-series expansions imply

$$\cosh u = \sum_{k=0}^{\infty} \frac{u^{2k}}{(2k)!} \leq \sum_{k=0}^{\infty} \frac{u^{2k}}{2^k k!} = e^{u^2/2},$$

where we used $(2k)! \geq 2^k k!$ and the nonnegativity of each term. Applying this inequality coordinate by coordinate yields

$$\begin{aligned}\mathbb{E}e^{tZ_i} &\leq \prod_{j=1}^m e^{t^2(v_i)_j^2/2} \\ &= e^{t^2\|v_i\|_2^2/2} \leq e^{t^2r^2/2}.\end{aligned}$$

Step 3: Bound the expected maximum. Write $M := \max_{i \in [N]} Z_i$. By Jensen's inequality for the convex function $z \mapsto e^{tz}$,

$$e^{t\mathbb{E}M} \leq \mathbb{E}e^{tM}.$$

For each realization of the signs, the largest of the positive numbers $e^{tZ_1}, \dots, e^{tZ_N}$ is at most their sum. Combining this fact with Step 2 gives

$$e^{t\mathbb{E}M} \leq \mathbb{E} \max_{i \in [N]} e^{tZ_i} \leq \sum_{i=1}^N \mathbb{E}e^{tZ_i} \leq Ne^{t^2r^2/2}.$$

Taking logarithms and dividing by $t > 0$, we obtain

$$m \mathbf{R}(A) = \mathbb{E}M \leq \frac{\log N}{t} + \frac{tr^2}{2} \quad \text{for every } t > 0.$$

The random variables Z_i need not be independent; only the coordinate signs σ_j were required to be independent in Step 2.

Step 4: Optimize the parameter and handle the boundary cases. If $r = 0$, all the centered vectors v_i equal zero. Thus $M = 0$, so $\mathbf{R}(A) = 0$ and the claimed bound holds with equality. This includes $N = 1$, because then $\bar{a} = a_1$.

Otherwise $r > 0$ and necessarily $N \geq 2$. The function

$$g(t) := \frac{\log N}{t} + \frac{tr^2}{2}$$

has derivative $g'(t) = -(\log N)/t^2 + r^2/2$ and second derivative $g''(t) = 2 \log N/t^3 > 0$. Its minimum over $t > 0$ is therefore attained at

$$t^* = \frac{\sqrt{2 \log N}}{r}.$$

Substituting this value into the preceding bound gives

$$m \mathbf{R}(A) \leq \frac{r \log N}{\sqrt{2 \log N}} + \frac{r \sqrt{2 \log N}}{2} = r \sqrt{2 \log N}.$$

Finally, dividing by m and substituting the definition of r proves

$$\mathbf{R}(A) \leq \max_{i \in [N]} \|a_i - \bar{a}\|_2 \frac{\sqrt{2 \log N}}{m}.$$

□